

Module 6. Cluster Analysis and Selected Methods

- ❑ Cluster Analysis: Basic Concepts
- ❑ Partitioning Methods
- ❑ Hierarchical Methods 
- ❑ Density-based and Grid-based Methods
- ❑ Advanced Methods
- ❑ Evaluation of Clustering
- ❑ Summary

Hierarchical Clustering: Basic Concepts

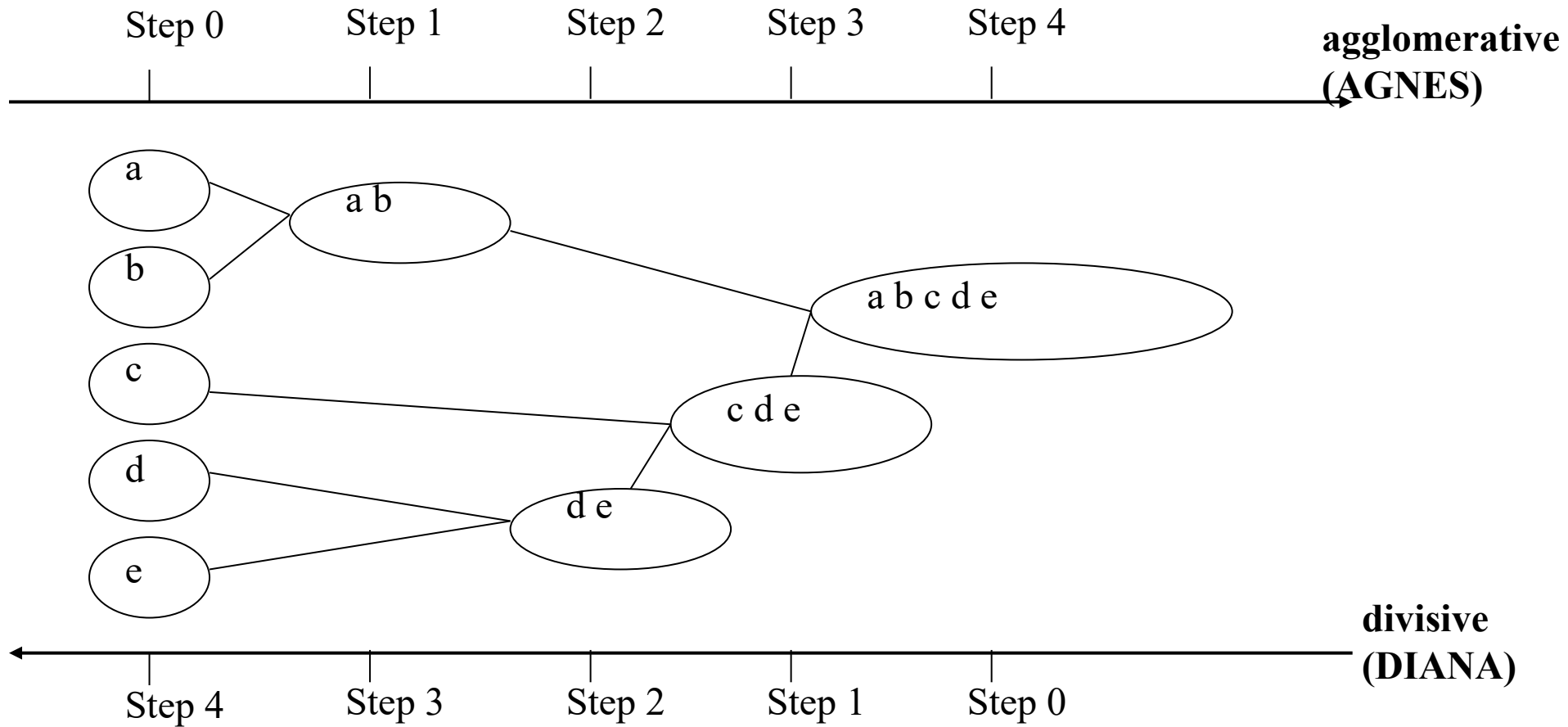
❑ Hierarchical clustering

- ❑ Generate a clustering hierarchy (drawn as a dendrogram)
- ❑ Not required to specify K , the number of clusters
- ❑ More deterministic
- ❑ No iterative refinement

❑ Two categories of algorithms

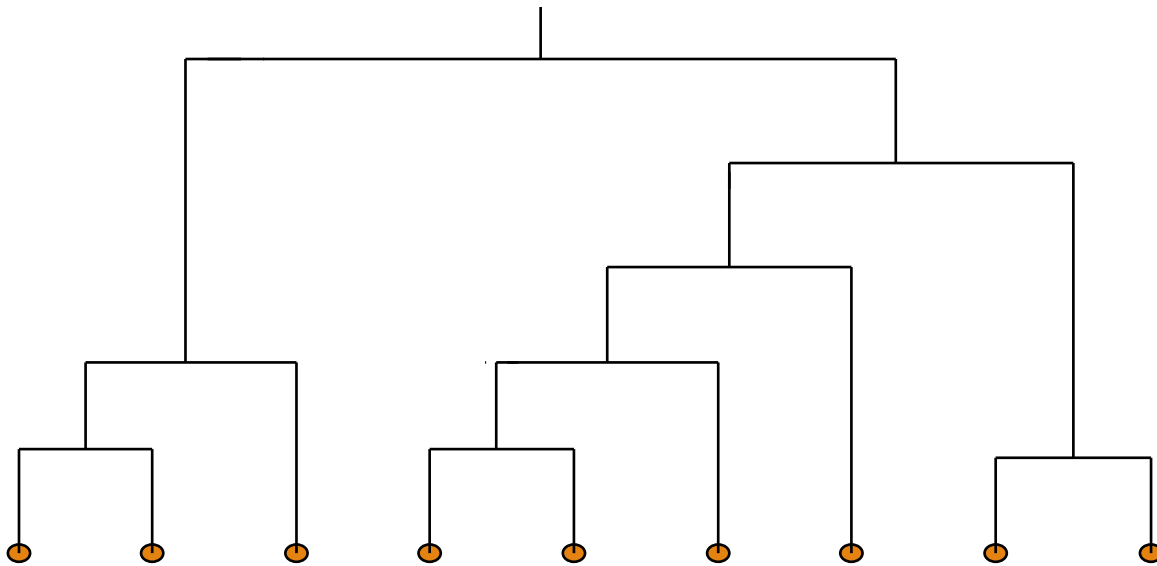
- ❑ Agglomerative: Start with singleton clusters, continuously merge two clusters at a time to build a bottom-up hierarchy of clusters
- ❑ Divisive: Start with a huge macro-cluster, split it continuously into two groups, generating a top-down hierarchy of clusters

Agglomerative vs. Divisive Clustering



Dendrogram: How Clusters are Merged

- ❑ Dendrogram: Decompose a set of data objects into a tree of clusters by multi-level nested partitioning
- ❑ A clustering of the data objects is obtained by cutting the dendrogram at the desired level, then each connected component forms a cluster



Hierarchical clustering generates a dendrogram (a hierarchy of clusters)